Master: Advance Mechatronic

PROJ 703

Eurobot 2026

SSR Team

Document: Systems Analysis and Preliminary Design Specification

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Contents

Section 1 : Introduction

The objective of the Eurobot contest is to create an autonomous vehicle capable of collecting hazelnut crates from the playing arena and transporting them to specific, pre-defined locations. These actions are governed by a strict set of rules and must be performed without human intervention.

Achieving this goal requires the vehicle to possess a specific set of functionalities. For example, it must be able to:

- Navigate the playing arena while avoiding both static obstacles and the opposing team's moving vehicle.
- Detect the locations of all hazelnuts and identify its target.
- Overcome various other technical challenges.

This document will define these challenges with the ultimate goal of defining the various systems required and its functionalities for the robot to compete successfully in the Eurobot contest.

This documentation pertains to the “Eurobot2026_Rules_BETA_0.9_EN” document. It is highly recommended to read the rules document first before continuing.

Section 2 : The Eurobot Actions

The Eurobot challenge includes 5 Actions where each action has its own description, rules and pointing criteria. Our objective in the SSR team is to be ready to participate in the main action which is called “LET’S KEEP THE HAZELNUTS WARM!”.

According to the “Eurobot2026_Rules_BETA_0.9_EN” document the action goal and constrains are as follow:

2.1 The Goal Of “LET’S KEEP THE HAZELNUTS WARM!” :

“The goal is to collect as many hazelnut crates as possible and deposit them in a pantry or in the squirrel nest “

2.2 The Constraints of The Action:

- A hazelnut crate is considered valid in a zone if all or part of its vertical projection is in this zone, and if it is in flat contact with the table.
- A hazelnut crate placed in the squirrel nest is valid for the team regardless of its color, and it cannot be stolen by the opposing team. It is possible to put a maximum of 6 crates in the nest, excess hazelnut crates will not be counted.
- A hazelnut crate placed in the pantries is valid for the team if its upper face is the team’s color or on a neutral face. There is no limit to the number of hazelnut crates placed in these zones. Hazelnut crates thus placed can be stolen by the opposing team.
- Each pantry brings an interest at the end of the match. For a zone to bring interest to a team, this team must have a majority of hazelnut crates of its color in the zone. In the event of a tie, there is no interest on the zone in question.
- A crate of rotten hazelnuts does not earn points for this action.
- An element still under control by a robot or a SIMA at the end of the match will not be counted.

Section 3 : SSR Team Objectives

Our main objectives regarding to “LET’S KEEP THE HAZELNUTS WARM!” action are:

- Our robot design should be simple and convenient with embedded gripping mechanisms.
- The robot should be able to grip and transport one hazelnut crates at a time.
- The robot should know the location of each and every hazelnut crate in the playing arena.
- The robot should be able to set a target and know how to move to it based on a set of criteria.
- The robot should define the most suitable location for the hazelnut crate which it is holding (nest or pantry).
- The robot should be able to steal a hazelnut crate from the opposite team.

Section 4 : General Requirements and Technical Challenges

Based on our team’s objectives there are multiple requirement and technical challenges and they are defined as follow:

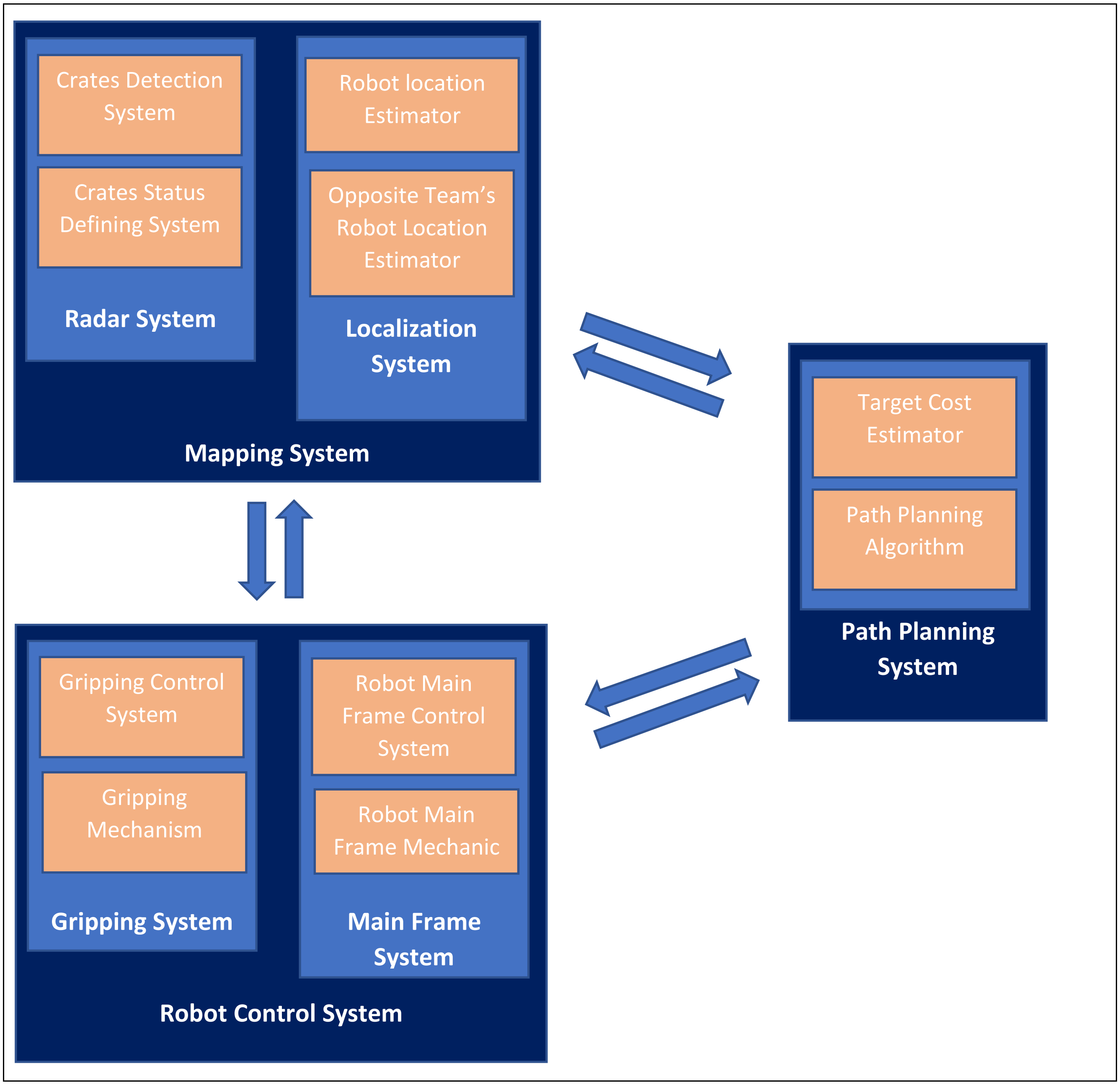
1. The robot location -robot frame in accordance to the base frame which is represented by the playing arena - should be known at any given time and continuously updated with reasonable accuracy.
2. The hazelnut crate’s locations and orientations -hazelnut crate’s frame- should be known at any giving time and continuously updated with reasonable accuracy.
3. The path planning algorithm should be simple robust and able to update the trajectory according to live map to avoid obstacles like opposite team moving robot and hazelnuts crates all over the playing arena.
4. The opposite team robot location should be known and updated regularly with estimated trajectory.
5. The robot mechanical design should be simple yet robust.
6. The gripping mechanism should be minimalist and act according to the hazelnut crates colors.
7. Hazelnut crates counts should be known in the nest since the nest can’t hold more than 6 crates and decide the pantries status if they are full or has empty spaces for additional crate.
8. An algorithm is needed to define the best destination -nest or pantries- for each crate outside those two destinations on the playing arena and calculate the cost of gripping and transport each one of them to help the robot be more efficient.

Section 5 : System Architecture

The analysis of the information presented in this document demonstrates that an onboard system in a mobile robot is incapable of fulfilling all the stated objectives. Therefore, the implementation of external support systems is necessary to direct the robot towards the most advantageous target through continuous, real-time match analysis and dynamic strategy adaptation.

This section outlines the foundational architecture of our proposed solution, designed to mitigate the identified challenges and fulfill the system requirements. As illustrated in Diagram 1, the architecture comprises three interconnected primary systems:

- Mapping System
- Path Planning System
- Robot Control System



- Diagram 1 System Architecture -

5.1 Mapping System

This system is tasked with gathering all necessary data from the playing arena to maintain up-to-date situational awareness and provides an interface for communicating these results to the Path Planning System.

This system is divided into two sub-systems which they are:

- Radar System
- Localization System

5.1.1 Radar System

The primary function of this subsystem is to determine the exact location and orientation of each box within the playing field, and then determine its status as a valid target or neutralized.

A crate is classified as a target if it is either outside any nest or pantry, or inside a pantry with the opposing team's color face-up. Otherwise, it is neutralized.

5.1.2 Localization System

his sub-system is dedicated to the real-time tracking and motion forecasting of all robotic agents—both ally and opponent—within the operational environment.

5.2 Path Planning System

The Path Planning System is responsible for selecting the optimal target within the playing arena based on cost metrics derived from the Mapping System. Subsequently, it computes an optimal trajectory for the robot to navigate towards the selected target. This system transmits the resulting trajectory to the Robot Control System and monitors the robot's execution progress. Acting as the central coordination module.

5.3 Robot Control System

It will be divided into two sub-systems:

5.3.1 Main Frame Movement Control Sub-system

This module is responsible for the precise execution of trajectory commands received from the Path Planning System. It simultaneously streams internal sensor data back to the Mapping System to enhance the accuracy of the robot's localized position.

5.3.2 Gripping System

This module is activated upon the successful completion of a movement command. Its operation is contingent on the color of the crate's upper face. After executing its gripping or release action, it signals the Main Frame Movement Control Sub-system to proceed with navigation to the next target location.